

MODULE	LEARNING OBJECTIVES	ACTIVITIES	ASSESSMENTS
Module 0 START HERE: Course Orientation	● 0.1 Configure Canvas notification settings and verify access to JupyterLab, Python, and the course data files ● 0.2 Examine course policies, grading criteria, and campus support resources to prepare for academic success ● 0.3 Compose a reflective learning plan connecting personal goals and prior experience to course learning outcomes	● Syllabus Review ● Course policy review ● Canvas navigation ● JupyterLab and Python tools review ● Academic Integrity ● AI use overview	● 0.5 Question & Discuss  Let's Get to Know Each Other (intro. discussion) ● 0.6 Confirm Your Course Readiness & Syllabus Review
Module 1 Introduction to Data Analysis Primary CLOs: ● I, II, VII	● 1.1 Explain the three main components of data science and how they work together in industry applications ● 1.2 Launch JupyterLab and create a functional Python notebook for data analysis ● 1.3 Use pandas to load a CSV dataset and perform basic DataFrame operations	● Read Sections: 1.1 - 1.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 1.6 Skills Lab 1A: Your First Data Analysis Workflow (Parts 1-3 and Questions & Analysis)
Module 2 Working with JupyterLab Primary CLOs: ● VII	● 2.1 Launch JupyterLab and create functional Python notebooks with code and Markdown cells ● 2.2 Use keyboard shortcuts and Tab completion to work efficiently in JupyterLab ● 2.3 Interpret error messages to identify and fix common Python coding errors	● Read Sections: 2.1 - 2.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 2.6 Skills Lab 2A: Create a Professional Analysis Notebook (Parts 1-3 and Q & A)
Module 3 Data Structures and Control Flow Primary CLOs: ● VII	● 3.1 Create and manipulate lists, tuples, and dictionaries to store and organize data collections ● 3.2 Implement conditional statements (if/elif/else) to control program flow based on different conditions ● 3.3 Use for loops and while loops to iterate over data structures and repeat operations efficiently	● Read Sections: 3.1 - 3.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 3.6 Skills Lab 3A: Building a Grade Analysis Tool (Parts 1-3 and Questions & Analysis) ● 3.7 Question & Discuss  Data Structures & Control Flow
Module 4 Working with pandas DataFrames Primary CLOs: ● II, IV	● 4.1 Create pandas DataFrames from CSV files and Python data structures, understanding the relationship between dictionaries/lists and DataFrame structure ● 4.2 Select specific columns and rows from DataFrames using bracket notation, .loc[], and .iloc[] indexing methods ● 4.3 Filter DataFrame rows using .query() and boolean indexing to extract data subsets based on conditions	● Read Sections: 4.1 - 4.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 4.6 Skills Lab 4A: City Population Analysis (Parts 1-3 and Questions & Analysis)

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Module 5 Data Exploration and Summary Statistics Primary CLOs: VII	5.1 Use <code>.info()</code>, <code>.describe()</code>, and <code>.value_counts()</code> to inspect DataFrame structure, generate statistical summaries, and identify data distributions 5.2 Calculate and interpret measures of central tendency (mean, median, mode) and measures of spread (range, variance, standard deviation) to characterize dataset distributions 5.3 Use <code>.groupby()</code> and aggregation methods to analyze data by categories and compare statistics across different subgroups	- Read Sections: 5.1 - 5.7 - Practice: Try It Yourself - Debug: Fix It  - Answer: Quick Checks - Recall: Retrieval Practice - Review, Further Reading, and Looking Ahead	5.6 Skills Lab 5A: Analyzing Sales Performance Data (Parts 1-3 and Questions & Analysis) 5.7 Question & Discuss  Libraries Assisting Data Analysis
Module 6 Data Cleaning and Preparation Primary CLOs: II	6.1 Identify missing data patterns using <code>.isnull()</code>, <code>.notnull()</code>, and <code>.info()</code> 6.2 Handle missing data appropriately using <code>.dropna()</code> and <code>.fillna()</code> 6.3 Convert data types, remove duplicates, and clean string data using pandas methods	- Read Sections: 6.1 - 6.7 - Practice: Try It Yourself - Debug: Fix It  - Answer: Quick Checks - Recall: Retrieval Practice - Review, Further Reading, and Looking Ahead	6.6 Skills Lab 6A: Cleaning Customer Survey Data (Parts 1-3 and Questions & Analysis)
Module 7 Data Visualization with matplotlib Primary CLOs: VI	7.1 Create line plots, bar charts, and scatter plots using <code>matplotlib.pyplot</code> 7.2 Customize plot appearance with titles, labels, colors, and styles 7.3 Select appropriate visualization types based on data characteristics and analytical goals	- Read Sections: 7.1 - 7.7 - Practice: Try It Yourself - Debug: Fix It  - Answer: Quick Checks - Recall: Retrieval Practice - Review, Further Reading, and Looking Ahead	7.6 Skills Lab 7A: Analyzing Temperature Trends (Parts 1-3 and Questions & Analysis) 7.7 Question & Discuss  Data Visualization
Module 8 Advanced Data Manipulation Primary CLOs: II	8.1 Sort DataFrames by single or multiple columns and create rankings to identify patterns in data 8.2 Create pivot tables, cross-tabulations, and reshape data between wide and long formats to summarize and prepare multidimensional data 8.3 Merge and concatenate multiple DataFrames to combine data from different sources	- Read Sections: 8.1 - 8.7 - Practice: Try It Yourself - Debug: Fix It  - Answer: Quick Checks - Recall: Retrieval Practice - Review, Further Reading, and Looking Ahead	8.6 Skills Lab 8A: Regional Sales Analysis Project (Parts 1-3 and Questions & Analysis)
Module 9 Introduction to Statistical Analysis Primary CLOs: VI, VII	9.1 Calculate and interpret correlation coefficients using <code>.corr()</code> method and scatter plots to quantify the strength and direction of relationships between numeric variables 9.2 Explain the logic of hypothesis testing including null hypotheses, alternative hypotheses, and p-values, and determine statistical significance at common alpha levels 9.3 Compare groups using independent samples t-tests and chi-square tests (<code>scipy.stats.chi2_contingency()</code>), interpreting the resulting p-values to judge statistical significance	- Read Sections: 9.1 - 9.7 - Practice: Try It Yourself - Debug: Fix It  - Answer: Quick Checks - Recall: Retrieval Practice - Review, Further Reading, and Looking Ahead	9.6 Skills Lab 9A: Statistical Analysis of Education Data (Parts 1-3 and Questions & Analysis) 9.7 Question & Discuss  Advanced Data Analysis

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Module 10 Functions and Code Reusability Primary CLOs: VII	● 10.1 Define and call functions with parameters and return values to perform reusable calculations ● 10.2 Write docstrings following professional documentation standards to explain function purpose and usage ● 10.3 Organize data analysis workflows into modular, maintainable functions	● Read Sections: 10.1 - 10.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 10.6 Skills Lab 10A: Building a Reusable Analysis Library (Parts 1-3 and Questions & Analysis)
Module 11 Working with External Data Sources Primary CLOs: III, IV, V	● 11.1 Read and write data in multiple file formats including CSV, Excel, and JSON with appropriate parameters ● 11.2 Use the requests library to fetch data from web APIs and convert responses into DataFrames ● 11.3 Explain ethical considerations and best practices for web scraping and database access	● Read Sections: 11.1 - 11.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review, Further Reading, and Looking Ahead	● 11.6 Skills Lab 11A: Multi-Source Data Integration Project (Parts 1-3 and Questions & Analysis) ● 11.7 Question & Discuss  The Future of Data Analytics (Overall Course Concepts)
Module 12 Ethics, Reproducibility, and Next Steps Primary CLOs: I, VII	● 12.1 Identify potential sources of bias in datasets and analysis processes, and propose mitigation strategies ● 12.2 Create reproducible analysis workflows using clear documentation, organized code, and version control ● 12.3 Explain career pathways in data analytics and identify resources for continued professional development	● Read Sections: 12.1 - 12.7 ● Practice: Try It Yourself ● Debug: Fix It  ● Answer: Quick Checks ● Recall: Retrieval Practice ● Review Questions ● Course Conclusion	● 12.6 Skills Lab 12A: Portfolio-Ready Analysis Project (Parts 1-3 and Questions & Analysis)